

Graph Theory

A graph is a mathematical structure used to represent a set of objects and the connections between them -

- a. The objects are called vertices (or nodes),
- b. The connections between them are called edges (or links).

Basic Concepts

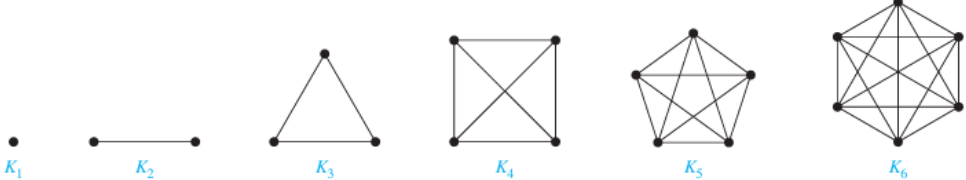
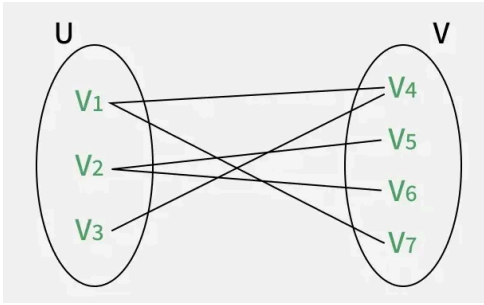
1. Degree of a Vertex: The number of edges incident on a vertex. In a directed graph, we distinguish between in-degree (incoming edges) and out-degree (outgoing edges).
2. Parallel Edges: Two or more edges that connect the same pair of vertices.
3. Pseudographs - Graphs that may include loops, and possibly multiple edges connecting the same pair of vertices or a vertex to itself, are sometimes called pseudographs.
4. Mixed graph - A graph with both directed and undirected edges is called a mixed graph.

Let $G = (V, E)$ be a graph with directed edges. Then

$$\sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v) = |E|.$$

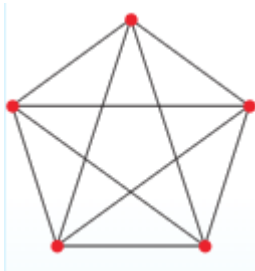
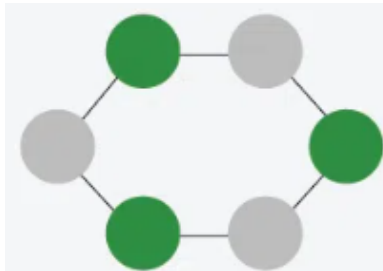
5.

Types of graph		
Directed (Digraph)	Directed Acyclic Graph (DAG)	It is a directed graph with no cycle. For a vertex v in DAG, there is no directed edge starting and ending with vertex v. The arrows go in one direction only (Directed), and you can't go in a circle or loop (Acyclic). Properties - 1. Directed edges – Every edge has a direction, from one vertex to another. 2. Acyclic – There are no directed cycles; you cannot start at a vertex and follow edge directions back to the same vertex. 3. Topological ordering exists – Every DAG has at least one topological ordering of its vertices. 4. At least one source vertex – A DAG always has at least one vertex with in-degree 0 (a source). 5. At least one sink vertex – A DAG always has at least one vertex with out-degree 0 (a sink). 6. No self-loops – A self-loop would form a cycle of length 1, so it is not allowed. 7. Partial order representation – A DAG can represent a partial ordering among elements. 8. Reachability is transitive – If there is a path from A to B and from B to C, then A precedes C in the dependency structure. 9. Can be traversed without revisiting vertices in a cycle – Since cycles do not exist, traversal algorithms avoid infinite loops caused by cycles. 10. Supports dynamic programming – Many optimization problems on DAGs can be solved efficiently because vertices can be processed in topological order.
	Tree	A tree is just a restricted form of graph. That is, it is a DAG with a restriction that a child can have only one parent.
Undirected	Connected	Connected Graph - A graph is connected if there is a path between every pair of vertices.

		- Minimum Edges: For a graph with n vertices, you need at least $n - 1$ edges to keep it connected - Vertex Degree: Vertices can have any degree (≥ 1)
	Complete	Complete Graph - A graph is complete if every single vertex is directly connected to every other vertex by an edge - A complete graph with n vertices is denoted as K_n, and it always contains exactly $n(n - 1)/2$ edges - Every vertex has the exact same degree: $n - 1$ The graphs K_n, for $n=1,2,3,4,5,6$, are displayed in the following figure - 
	Biconnected	A connected graph that cannot be broken down into any further pieces by the deletion of any vertex. It is a graph with no articulation point.
	Bipartite	A Bipartite Graph (sometimes called a bigraph) is a special type of graph in graph theory where the vertices (nodes) can be divided into two completely separate, independent sets, say U and V, such that every single edge in the graph connects a vertex from U to a vertex from V.  Key Characteristics of a Bipartite Graph - 1. Vertex Partitioning - The total set of vertices V_G can be partitioned into two disjoint sets, U and V ($U \cup V = V_G$ and $U \cap V = \emptyset$). 2. No Odd-Length Cycles - A graph is bipartite if and only if it contains no odd cycles. It can have even cycles (cycles of length 4, 6, 8, etc.). If you find a triangle (cycle of length 3) or a pentagon (cycle of length 5) anywhere in the graph, it is strictly not bipartite. 3. Two-Colorability - The chromatic number of a bipartite graph is always ≤ 2. This means you can color the vertices using at most two colors such that no adjacent vertices have the same color. 4. Every subgraph of a bipartite graph is also a bipartite graph. 5. Maximum Matchings (Hall's Marriage Theorem) - According to Hall's Marriage Theorem, a perfect matching from set U to set V exists if and only if for every subset of U, the number of its neighbors in V is greater than or equal to the size of that subset. 7. Number of Edges - $0 \leq \text{edges} \leq m * n$ while m is the size of set U and n is size of V.

Every complete graph is automatically a connected graph. Not every connected graph is a complete graph.

For example, a graph where 5 vertices are arranged in a straight line (A - B - C - D - E) is fully connected because you can get from A to E, but it is not complete because A doesn't connect directly to C, D, or E.

A Connected Graph is a graph in which there is a path between every pair of vertices. In other words, starting from any vertex, you can reach every other vertex by following the edges of the graph. For example, if $A \rightarrow B \rightarrow C \rightarrow D$, the graph is connected because every vertex can be reached from every other vertex.	A Complete Graph is an undirected graph in which every pair of distinct vertices is connected by a unique edge. This means the graph contains the maximum possible number of edges for its number of vertices. It is usually denoted as K_n, where n is the number of vertices.	A Bipartite Graph is a graph whose vertices can be divided into two independent, disjoint sets (let's call them U and V) such that every edge connects a vertex in U to a vertex in V. No edges exist between vertices within the same set.
		

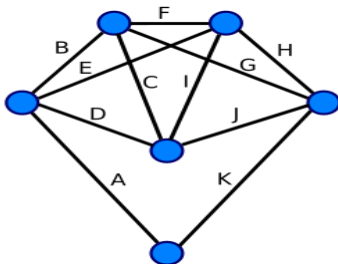
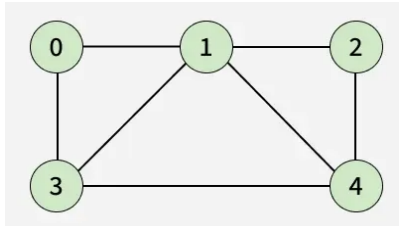
A **Hamiltonian Graph** is a graph that contains a Hamiltonian Cycle - a closed path that visits every vertex exactly once and returns to the starting vertex.

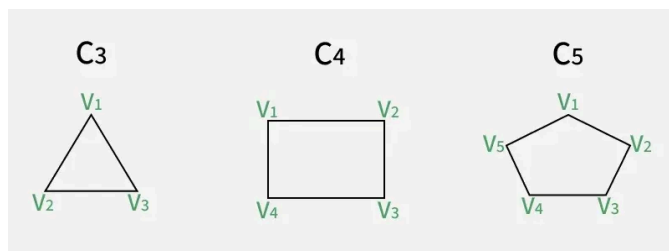
Finding a Hamiltonian cycle is an **NP Complete** problem, meaning there's no known efficient solution for all graph types, but solutions exist for smaller or specific types.

- **Dirac's Theorem:** If a simple graph has n vertices (where $n \geq 3$), and every vertex is connected to at least $n/2$ other vertices, the graph is Hamiltonian.
- **Ore's Theorem:** If a simple graph has n vertices ($n \geq 3$) and the sum of connections (degrees) of any two non-connected vertices is at least n , the graph is Hamiltonian.

A **Eulerian Graph** has an Eulerian circuit. All vertices have an even degree, and the path starts and ends at the same vertex. **Semi-Eulerian Graph:** Has an Eulerian trail. It visits every edge exactly once but starts and ends at different vertices. **This happens if and only if the graph has exactly two vertices with an odd degree.**

A connected graph is considered **Eulerian** if and only if every vertex in the graph has an even degree (meaning an even number of edges are connected to each vertex).

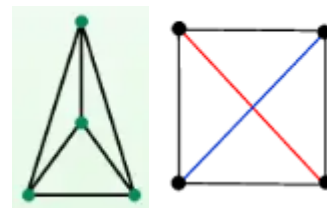
Feature	Eulerian Graph	Hamiltonian Graph
Focus	Traverses every edge exactly once.	Visit every vertex exactly once.
Repeat?	Vertices can be visited multiple times.	Edges do not all need to be used.
Core Condition	Every vertex must have an even degree (for a circuit).	No simple condition exists (NP Complete problem)
Origin	Seven Bridges of Königsberg problem	Around the World game (Sir William Hamilton)
Example		



Cycle graph: It is a connected graph where each vertex has degree 2, forming a single closed loop without any branches or endpoints. This graph contains at least 3 vertices. Suppose a graph has the following vertices: $v_1, v_2, v_3, \dots, v_n$. This graph will be a cycle graph if it has edges as follows: $(v_1, v_2), (v_2, v_3), (v_3, v_4), \dots, (v_{n-1}, v_n), (v_n, v_1)$.

Planar Graph - A planar graph is a graph that can be drawn on a two-dimensional plane (like a sheet of paper) in such a way that no two edges cross each other. In other words, the edges only intersect at their endpoints (the vertices).

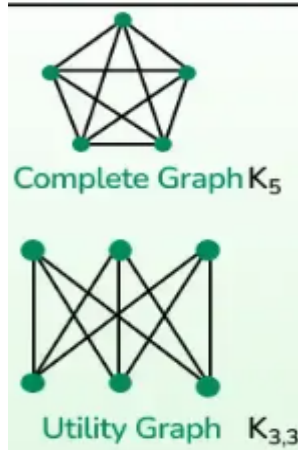
A graph that cannot be drawn without at least one pair of edges crossing is called a non-planar graph. A classic example of a planar graph is the complete graph with 4 vertices (K_4), where every vertex is connected to every other vertex.



Key properties -

- Planar graphs cannot have an infinite number of edges; they are relatively sparse. For any simple connected planar graph with $V \geq 3$:
 - The number of edges will never exceed $3V - 6$ (i.e., $E \leq 3V - 6$).
 - If the graph contains no triangles (no cycles of length 3), the edge limit drops even further to $E \leq 2V - 4$.
- For any connected planar graph, there is a strict mathematical relationship between the number of Vertices (V), Edges (E), and Faces (F). This is known as **Euler's Formula**: $V - E + F = 2$.
For example, in the above K_4 graph, $V = 4$ and $E = 6$ then $4 - 6 + F = 2 \Rightarrow F = 4$.
- Five-Color Theorem** - Every planar graph can have its vertices colored using at most four colors such that no two adjacent vertices share the same color.
- Kuratowski's Theorem (How to detect Non-Planarity)** - Polish mathematician Kazimierz Kuratowski proved that a graph is non-planar if and only if it contains a subgraph that is a structural variant of one of these two specific graphs:
 - K_5** : The complete graph with 5 vertices (a star inside a pentagon).
 - $K_{3,3}$** : The complete bipartite utility graph (3 houses, 3 utilities, where every house connects to every utility).

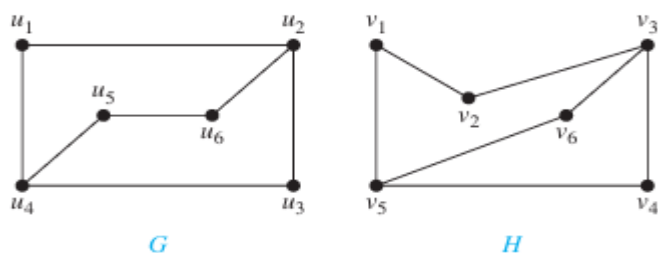
Non-Planar



Isomorphism of graph - The simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are isomorphic if there exists a one-to-one and onto function f from V_1 to V_2 with the property that a and b are adjacent in G_1 if and only if $f(a)$ and $f(b)$ are adjacent in G_2 , for all a and b in V_1 . Such a function f is called an isomorphism.

- Two simple graphs that are not isomorphic are called non-isomorphic.

In other words, when two simple graphs are isomorphic, there is a one-to-one correspondence between vertices of the two graphs that preserves the adjacency relationship. Isomorphism of simple graphs is an equivalence relation.



Example 01 - Determine whether the graphs shown in following figure are isomorphic

Criteria	G	H
Number of edges	7	7
Number of vertices	6	6
Nodes with degree two	4	4
Nodes with degree three	2	2
Adjacency matrix	$A_G = \begin{matrix} & \begin{matrix} u_1 & u_2 & u_3 & u_4 & u_5 & u_6 \end{matrix} \\ \begin{matrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$	$A_H = \begin{matrix} & \begin{matrix} v_6 & v_3 & v_4 & v_5 & v_1 & v_2 \end{matrix} \\ \begin{matrix} v_6 \\ v_3 \\ v_4 \\ v_5 \\ v_1 \\ v_2 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$
So we can say that G and H are isomorphics		

Chromatic number of a graph - The chromatic number of a graph is the least number of colors needed for coloring a graph.

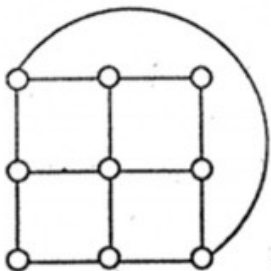
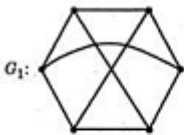
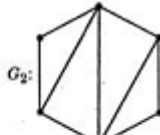
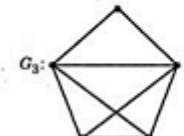
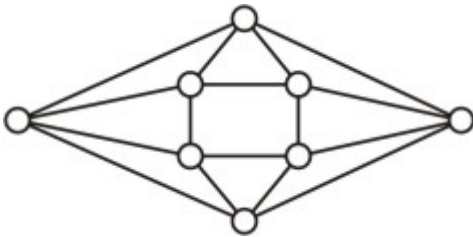
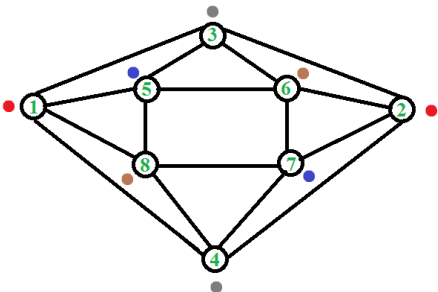
The chromatic number of a graph G is denoted by $\chi(G)$. (Here χ is the Greek letter chi.)

THE FOUR COLOR THEOREM - The chromatic number of a planar graph is no greater than four. Non-planar graphs can have arbitrarily large chromatic numbers.

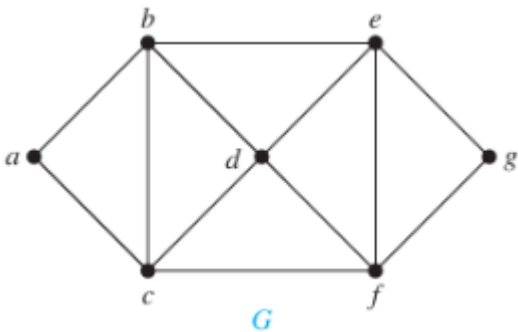
A graph is called planar if it can be drawn in the plane without any edges crossing (where a crossing of edges is the intersection of the lines or arcs representing them at a point other than their common endpoint). Such a drawing is called a planar representation of the graph

Graph Type	Chromatic Number $\chi(G)$	Key Structural Reason
Empty Graph	1	No edges exist.
Bipartite Graph	2	Vertices can be cleanly split into two independent sets.
Even Cycle (C_{2k})	2	Alternating two colors completes the loop perfectly.
Odd Cycle (C_{2k+1})	3	The final closing vertex is adjacent to both starting colors.
Complete Graph (K_n)	n	Every vertex requires a totally unique color.
Planar Graph	≤ 4	Edges do not cross on a 2D plane (Four Color Theorem).

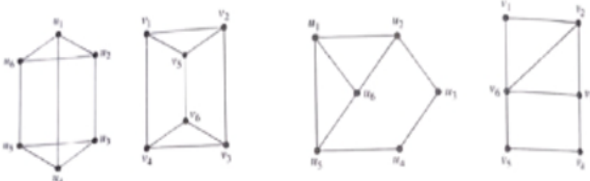
1	The minimum number of colours that is sufficient to vertex-colour any planar graph is _	4
2	Let G be a connected planar graph with 10 vertices. If the number of edges on each face is three, then the number of edges in G is _	$V = 10$, As, number of edges on each face is three, so $2E = 3F \Rightarrow F = 2E / 3$,

		We know, $10 - E + 2E / 3 = 2$ $\Rightarrow 30 - E = 6$ $\Rightarrow E = 24$
3	What is the chromatic number of the following graph? 	3
4	Let G be a simple connected planar graph with 13 vertices and 19 edges. Then, the number of faces in the planar embedding of the graph is	Given, $V = 13$ and $E = 19$ We know that, $V - E + F = 2$ $\Rightarrow 13 - 19 + F = 2 \Rightarrow F = 8$
5	Which one of the following graphs is NOT planar? G_1:  G_2:  G_3:  G_4: 	G_1
6	The minimum number of colours required to colour the following graph, such that no two adjacent vertices are assigned the same colour, is 	
7	The minimum number of colours required to colour the vertices of a cycle with n nodes in such a way that no two adjacent nodes have the same colour is -	$n - 2\lfloor n/2 \rfloor + 2$

What is the chromatic number of following graphs G and H?

 G	The chromatic number of G is at least three, because the vertices a,b, and c must be assigned different colors.
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	<i>G</i>
<i>H</i>	H has a chromatic number equal to 4. <i>H</i>
Given an undirected graph $G(V, E)$, what is the maximum number of edges it can have?	$v * (v - 1) / 2$
How many edges are there in a graph with 10 vertices each of degree six?	According to Handshake Theorem, given an undirected graph, $G = (V, E)$ with m edges, then, $2m = \sum_{v \in V} \deg(v).$ Here, 10 vertices each with degree six, then $2m = 6 * 10 = 60$ $\Rightarrow m = 60 / 2 \Rightarrow m = 30$ So there are 30 edges
Are the graphs G and H displayed in the following figure bipartite? <i>G</i> <i>H</i>	Graph G is bipartite because its vertex set is the union of two disjoint sets, which are $\{a, b, d\}$ and $\{c, e, f, g\}$, and each edge connects a vertex in one of these subsets to a vertex in the other subset. To be bipartite it is necessary that every vertex in $\{a, b, d\}$ be adjacent to every vertex in $\{c, e, f, g\}$. For instance, b and g are not adjacent. Graph H is not bipartite because its vertex set can't be partitioned into two subsets so that edges do not connect two vertices from the same subset.
The chromatic index of a bipartite graph with at least one edge is __ [49th BCS (CSE)] a) 1 b) 2 c) 3 d)	Ans - b.
Define a graph and explain the concepts of degree, path, cycle and connectedness in an undirected graph. [47th BCSE (ICT) - Written]	A graph is a mathematical structure used to model pairwise relations between objects. An undirected graph $G = (V, E)$ consists of two sets: V - A non-empty set of vertices and E - A set of edges, where each edge connects a pair of vertices.

	The degree of a vertex in an undirected graph is the total number of edges connected to (or incident upon) it. It is typically denoted as $\deg(v)$ for a given vertex v. A path in a graph is a sequence of vertices where each consecutive pair of vertices is connected by an edge, and no vertices (or edges) are repeated. A cycle is a specific type of path that starts and ends at the exact same vertex, forming a closed loop. A graph that contains at least one cycle is called a cyclic graph, while a graph with absolutely no cycles is known as an acyclic graph. Connectedness describes whether a graph acts as a single, unified structure. An undirected graph is considered connected if there is at least one valid path between every possible pair of vertices in the graph. If the graph is split into two or more distinct, non-overlapping parts where you cannot travel from a vertex in one part to a vertex in another, the graph is disconnected.
Differentiate between Eulerian and Hamiltonian graphs with examples. [47th BCSE (ICT) - Written]	
Determine whether the following pair of graphs are isomorphic, and justify your answer in one sentence [Officer (IT) - 2025] 	

Propositional Logic

First Order or Predicate Logic

Aspect	Propositional Logic (PL)	First-Order Logic (FOL)
Definition	Propositional Logic is a knowledge representation method that represents facts as simple statements (propositions) that can be either True or False .	First-Order Logic, also called Predicate Logic , is an advanced logic system that represents objects, properties, relationships, and rules using predicates, variables, and quantifiers.
Basic Unit	The basic unit is a proposition/sentence .	The basic unit is a predicate involving objects and relationships.
Representation	Represents facts using symbols such as P, Q, R.	Represents facts using predicates, variables, constants, and functions .
Expressiveness	Less expressive; can represent simple facts only.	More expressive; can represent complex relationships and general rules.

Relationships	Cannot describe relationships	Can represent relationships
Variables	Does not use variables.	Uses variables such as x, y, and z.
Knowledge Representation	Represents knowledge as independent statements.	Represents knowledge using facts, rules, and relationships.
Reasoning Ability	Provides limited reasoning ability.	Provides powerful reasoning and inference capabilities.
Complexity	Simple and easier to implement.	More complex and requires greater computational resources.
Handling General Rules	Cannot express general rules effectively.	Can express general rules and universal statements.
Statement - "All humans are mortal. Socrates is a human. Therefore, Socrates is mortal."	Represented as separate facts: P = "All humans are mortal" Q = "Socrates is human" R = "Socrates is mortal"	As $\forall x(\text{Human}(x) \rightarrow \text{Mortal}(x))$ Human(Socrates) Therefore, Mortal(Socrates)

Syntax of First Order Logic -

- Terms** represent objects or entities within the domain of discourse. In AI, terms can correspond to real-world entities, such as objects, individuals, or abstract concepts.
 - Constants: Represent specific entities such as "John", "Apple", or "5".
 - Variables: Act as placeholders for objects such as "x", "y".
 - Functions: Map objects to other objects, e.g., "Age(John)", "Parent(x)".
- Predicates** express properties, relations, or conditions that hold between objects. They describe the state of the world or assert facts about entities within the domain.
 - IsHuman(x) means x is a human
 - IsParent(x, y) means x is a parent of y
- Quantifiers** in first-order logic allow for the specification of statements about the entirety or existence of objects within the domain.
 - Universal quantifiers ($\forall$):** States that a property holds for all objects, e.g., $\forall x P(x) \rightarrow$ "P(x) is true for every x"
 - Existential quantifiers ($\exists$):** Statements that hold for at least one object, e.g., $\exists x P(x) \rightarrow$ "There exists an x such that P(x) is true"
- Logical connectives** are used to combine or modify simple statements into complex ones. They help express logical relationships and constraints in AI knowledge representation.
 - Conjunction ($\wedge$) - If P(x) represents "x is red" and Q(x) represents "x is round", then $P(x) \wedge Q(x)$ represents "x is red and round"
 - Disjunction ($\vee$) - If P(x) represents "x is a cat" and Q(x) represents "x is a dog", then $P(x) \vee Q(x)$ represents "x is either a cat or a dog".
 - Implication ($\rightarrow$) - If P(x) represents "x is a mammal" and Q(x) represents "x produces milk", then $P(x) \rightarrow Q(x)$ represents "if x is a mammal, then it produces milk".
 - Negation ($\neg$) - If P(x) represents "x is intelligent", then $\neg P(x)$ represents "x is not intelligent".

Example of First Order Logic (FOL) Representation		
1	You can always fool some of the people	Meaning: There exists a group of people who can always be fooled.

		Let, $\text{People}(x) \rightarrow x$ is a person and $\text{Fool}(y, x) \rightarrow y$ can fool x FOL representation: $\exists x (\text{People}(x) \wedge \forall y \text{Fool}(y, x))$
2	All purple mushrooms are poisonous	Meaning: For every object x, if x is a purple mushroom, then x is poisonous. Let, $\text{PurpleMushroom}(x) \rightarrow x$ is a purple mushroom and $\text{Poisonous}(x) \rightarrow x$ is poisonous FOL representation: $\forall x (\text{PurpleMushroom}(x) \rightarrow \text{Poisonous}(x))$

Syntax of Propositional Logic -

Propositional logic (also called sentential logic) is a branch of formal logic that deals with propositions, statements that are either true or false, and the logical relationships between them, using connectives like "and," "or," "not," and "if...then."

Two main types of propositions -

1. Atomic Propositions: A simple statement without logical connectives that cannot be broken down further.
2. Compound Propositions: A statement formed by combining atomic propositions using logical connectives like AND, OR, or NOT.

Tautologies - A tautology is a proposition that is **true** regardless of the truth values assigned to its component propositions.

Contradictions - A contradiction is a proposition that is **false** regardless of the truth values assigned to its component propositions.

Contingencies - A contingency is a proposition that may not be necessarily **true or false**, it may be true in some situations but false in others.

Propositional Logic Representation					
Negation ($\neg$)	Not				
		p	$\neg p$		
		T	F		
		F	T		
Conjunction ($\wedge$)	And				
		p	q	$p \wedge q$	
		T	T	T	
		T	F	F	
		F	T	F	
		F	F	F	

Disjunction ($\vee$)	Or	<table><tr><th>p</th><th>q</th><th>$p \vee q$</th></tr><tr><td>T</td><td>T</td><td>T</td></tr><tr><td>T</td><td>F</td><td>T</td></tr><tr><td>F</td><td>T</td><td>T</td></tr><tr><td>F</td><td>F</td><td>F</td></tr></table>	p	q	$p \vee q$	T	T	T	T	F	T	F	T	T	F	F	F	
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T	T	T																
T	F	F																
F	T	F																
F	F	T																

Few examples and PYQs	
What are the strings in the regular sets specified by the regular expressions 10^* , $(10)^*$, $0 \cup 01$, $0(0 \cup 1)^*$, and $(0^*1)^*$?	10^* - The set of bit strings consisting of a 1 followed by zero or more 0s. $(10)^*$ - The set of bit strings formed by repeating the block 10 zero or more times $0 \cup 01$ - A finite set containing exactly two specific strings $0(0 \cup 1)^*$ - The set of all bit strings that begin with a 0 $(0^*1)^*$ - The set of all bit strings that do not end with a 0. In other words, it contains the empty string and all bit strings that end in 1
Find a regular expression that	a) $(00 \cup 01 \cup 10 \cup 11)^*$ or $((0 \cup 1)(0 \cup 1))^*$

specifies each of these sets: (a) the set of bit strings with even length (b) the set of bit strings ending with a 0 and not containing 11 (c) the set of bit strings containing an odd number of 0s	b) $(0 \cup 10)^* 0$ c) $1^* (01^* 01^*)^* 01^*$																																								
Is it tautology contradiction or contingency $P \rightarrow q = \neg p \rightarrow \neg q$ [ISTCL AP - 2026]	<table><tr><th>P</th><th>Q</th><th>$P \rightarrow Q$</th><th>P'</th><th>Q'</th><th>$P' \rightarrow Q'$</th><th>$P \rightarrow Q = P' \rightarrow Q$</th></tr><tr><td>T</td><td>T</td><td>T</td><td>F</td><td>F</td><td>T</td><td>T</td></tr><tr><td>T</td><td>F</td><td>F</td><td>F</td><td>T</td><td>T</td><td>F</td></tr><tr><td>F</td><td>T</td><td>T</td><td>T</td><td>F</td><td>F</td><td>F</td></tr><tr><td>F</td><td>F</td><td>T</td><td>T</td><td>T</td><td>T</td><td>T</td></tr></table> It is a Contingency as it contains both True and False logics.	P	Q	$P \rightarrow Q$	P'	Q'	$P' \rightarrow Q'$	$P \rightarrow Q = P' \rightarrow Q$	T	T	T	F	F	T	T	T	F	F	F	T	T	F	F	T	T	T	F	F	F	F	F	T	T	T	T	T					
P	Q	$P \rightarrow Q$	P'	Q'	$P' \rightarrow Q'$	$P \rightarrow Q = P' \rightarrow Q$																																			
T	T	T	F	F	T	T																																			
T	F	F	F	T	T	F																																			
F	T	T	T	F	F	F																																			
F	F	T	T	T	T	T																																			
Show that $p \leftrightarrow q$ and $(p \wedge q) \vee (\neg p \wedge \neg q)$ are logically equivalent [Directorate of Passport - AP - 2026]	<table><tr><th>p</th><th>q</th><th>$\neg p$</th><th>$\neg q$</th><th>$p \leftrightarrow q$</th><th>$p \wedge q$</th><th>$\neg p \wedge \neg q$</th><th>$(p \wedge q) \vee (\neg p \wedge \neg q)$</th></tr><tr><td>T</td><td>T</td><td>F</td><td>F</td><td>T</td><td>T</td><td>F</td><td>T</td></tr><tr><td>T</td><td>F</td><td>F</td><td>T</td><td>F</td><td>F</td><td>F</td><td>F</td></tr><tr><td>F</td><td>T</td><td>T</td><td>F</td><td>F</td><td>F</td><td>F</td><td>F</td></tr><tr><td>F</td><td>F</td><td>T</td><td>T</td><td>T</td><td>F</td><td>T</td><td>T</td></tr></table> Hence, the statement is logically equivalent.	p	q	$\neg p$	$\neg q$	$p \leftrightarrow q$	$p \wedge q$	$\neg p \wedge \neg q$	$(p \wedge q) \vee (\neg p \wedge \neg q)$	T	T	F	F	T	T	F	T	T	F	F	T	F	F	F	F	F	T	T	F	F	F	F	F	F	F	T	T	T	F	T	T
p	q	$\neg p$	$\neg q$	$p \leftrightarrow q$	$p \wedge q$	$\neg p \wedge \neg q$	$(p \wedge q) \vee (\neg p \wedge \neg q)$																																		
T	T	F	F	T	T	F	T																																		
T	F	F	T	F	F	F	F																																		
F	T	T	F	F	F	F	F																																		
F	F	T	T	T	F	T	T																																		

Pigeonhole Principle

The pigeonhole principle states that *if n items are put into m containers, with $n > m$, then at least one container must contain more than one item.*

Suppose that a flock of 13 pigeons flies into a set of 12 pigeonholes to roost.

At least one of these 12 pigeonholes must have at least two pigeons in it. At least one of these 12 pigeonholes must have at least two pigeons in it.

If we must put $N * k + 1$ or more pigeons into N pigeon holes, then some pigeon hole must contain $K + 1$ or more pigeons.		
1	If you have 10 black socks and 12 blue socks mixed in a drawer, how many socks must you pick to guarantee a matching pair?	Number of colors = 2 For making a pair you have to take at least $2 + 1 = 3$ socks.
2	If there are 100 students, how many students possibly share the same birth month?	For 100 students and 12 months, at least $\lceil 100/12 \rceil + 1 = \lceil 8.33 \rceil + 1 = 8 + 1 = 9$ students share a birth month.

3	A bag contains 10 red marbles, 10 white marbles and 10 blue marbles. What is the minimum number of marbles you have to choose randomly from the bag to ensure that we get 4 marbles of the same color?	No. of colors (pigeonholes) $n = 3$ No. of marbles to pick (pigeons) $K+1 = 4$ To choose 3 marbles of same color, we have to choose $3 * 3 = 9$ marbles. According to the pigeonhole principle, we may require a minimum $9 + 1 = 10$ marbles to pick from the bag to get 4 marbles of the same color.
4	In a computer science department, a student club can be formed with either 10 members from first year or 8 members from second year or 6 from third year or 4 from final year . What is the minimum no. of students we have to choose randomly from the department to ensure that a student club is formed?	Let, students from first year be $q1 = 10$, students from second year be $q2 = 8$, students from third year be $q3 = 6$, students from final year be $q4 = 4$. So total possible options are $n = 4$ Therefore the minimum number of students required to ensure department club to be formed is $10 + 8 + 6 + 4 - 4 + 1 = 25$
5	Ten students solved a total of 35 problems in a math olympiad. Each problem was solved by exactly one student. There is at least one student who solved exactly one problem, at least one student who solved two problems. and at least one student who solved exactly three problems. Prove that there is also at least one student who solved at least five problems .	Given, 3 students solved $1 + 2 + 3 = 6$ problems, so the remaining 7 students (let pigeonholes, $n = 7$) solved $35 - 6 = 29$ problems. So according to pigeonhole method, we got, $n * k + 1 = 29 \Rightarrow k * 7 = 28 \Rightarrow k = 4$ So some student may have solved $k + 1 = 4 + 1 = 5$ problems

Counting Principle

Addition Principle - *Example: If you have 3 shirts and 4 pants, there are $3+4 = 7$ ways to choose either a shirt or a pant.*

Multiplication Principle - *Example: If you have 3 shirts and 4 pants, there are $3 \times 4 = 12$ ways to choose a shirt and a pant.*

1	In how many ways can you choose 4 books from a shelf of 12 books?	4 Books from the shelf of 12 books can be chosen in ${}^4C_{12} = {}^{12}C_4 = 12! / 4! * (12-4)! = (12 \times 11 \times 10 \times 9) / (4 \times 3 \times 2 \times 1) = 495$ number of ways.
2	How many ways can you arrange the letters in the word "MATHEMATICS" such that the two M's are next to each other?	Treat two 'M's as a single unit. This reduces the problem to arranging the units: "MM", A, T, H, E, A, T, I, C, S (10 units). Here, A: 2 and T: 2 and other letters appear once. So total arrangement $= 10! / 2! \times 2! = 3628800 / 4 = 907200$
3	How many 5 digit numbers can be formed such that the digit 3 appears exactly twice?	We need to choose 2 positions out of 5 for the digit 3 which can be done in ${}^5C_2 = 5! / 2! * (5-2)! = 10$

		The remaining 3 positions can be filled with any digits from 0 to 9 except for 3. Since we are forming 5-digit numbers, the first digit cannot be 0. So, the first digit can be any digit from 1 to 9 except 3. That gives us 8 choices for the first digit. The other two positions can be any digit from 0 to 9 except 3. That gives us 9 choices for each of these two positions. For the first position we have 8 choices. For the second position we have 9 choices. For the third position we have 9 choices. Therefore, the number of ways to fill these remaining three positions is: $8 \times 9 \times 9 = 648$ Now, possible arrangement can be $10 \times 648 = 6480$
4	How many distinct ways are there to arrange 10 people in a row if exactly 3 people must be together in one block?	To solve this problem, we need to treat the block of 3 people who must be together as a single unit. The 3 people within a block can be arranged among themselves in $3!$ ways. [$3! = 6$] Since the 3 people are treated as one unit, we now have 8 units to arrange (the block plus the other 7 individual people). The 8 units can be arranged in $(8!)$ ways. [$8! = 40320$] So possible ways $8! \times 3! = 40320 \times 6 = 241920$
5	How many words with 2 different vowels and 2 different consonants can be formed from the alphabet?	The English alphabet contains 26 letters including 5 vowels and 21 consonants. Step 1: Ways of selecting 2 different vowels out of 5 vowels, $C(5, 2) = 5! / (2! \times (5 - 2)!) = 5! / (2! \times 3!) = 10$ Step 2: Ways of selecting 2 different consonants out of 21 consonants, $C(21, 3) = 21! / (2! \times (21 - 2)!) = 21! / (3! \times 19!) = 210$ Hence, combinations of 2 different vowels and 2 different consonants is $10 \times 210 = 2100$ Step 3: Ways arranging or forming 2 vowels and 2 consonants out of 4 letters, $P(4, 4) = 4! / (4 - 4)! = 4! / 0! = 4! = 24$ Therefore, the total number of ways is $2100 \times 24 = 54000$
6	How many different bit strings of length seven are there?	As the string consists of either 0 or 1, so we have 2 options and the length of the string is 7, so we have 7 positions. So total possible bit string is $2^7 = 128$
7	How many different license plates can be made if each plate	3 uppercase English letters may contain $26 * 26 * 26 = 26^3$ 3 digits may contain 0 to 9, which is $10 * 10 * 10 = 10^3$

	contains a sequence of three uppercase English letters followed by three digits (and no sequences of letters are prohibited, even if they are obscene)?	$So\ total\ possible\ license\ plates\ can\ be = 26^3 * 10^3$
8	The North American Numbering Plan (NANP) specifies the format of telephone numbers in the U.S., Canada, and many other parts of North America. Telephone number in this plan consists of 10 digits, which are split into a three digit area code, a three-digit office code, and a four-digit station code. To specify the allowable format, let X denote a digit that can take any of the values 0 through 9, let N denote a digit that can take any of the values 2 through 9, and let Y denote a digit that must be a 0 or a 1. In the old plan, the formats of the area code, office code, and station code are telephone numbers have the form NXX-NXX-XXXX. How many different North American telephone numbers are possible under the old plan and under the new plan?	<i>For N we have 2 to 9, total 8 digits. For X we have 0 to 9, total 10 digits. For Y we have 0 and 1, total 2 digits.</i> <i>Given code NXX-NYX-XXXX has Area code NXX which is $8 * 10 * 10 = 800$ Office code NYX which is $8 * 2 * 10 = 160$ Station code XXXX which is $10^4 = 10000$</i> <i>So, total telephone numbers possible is $800 * 160 * 10000 = 1280000000$</i>
9		
10		

Function and Relation

A **function** in mathematics f is defined as $y = f(x)$, where x is the input value, and for each input value of x , we get a unique value of y .

Example: $y = f(x) = 3x + 4$. This is a linear function. • If $x = 2$: $f(2) = 3(2) + 4 = 6 + 4 = 10$. So, the output is 10. • <i>Domain: All real numbers.</i> • <i>Range: All real numbers.</i>	Example: $y = f(x) = \sin(x) + 3$ This is a trigonometric function. • If $x = \pi/2$: $f(\pi/2) = \sin(\pi/2) + 3 = 1 + 3 = 4$. The output is 4. • Domain: All real numbers. • Range: $[2, 4]$.
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Domain and Range of a Function - The domain and range of a function are fundamental concepts in mathematics.

- Domain: The set of all possible input values (x-values) for which the function is defined.
- Range: The set of all possible output values (y-values) that the function can produce.

The **Domain and Range** of a function are the input and output value of a function, respectively.

For example: $f(x) = x^2$

- The domain is all real numbers (R) because any real number can be squared.
- The range is all non-negative real numbers ($[0, \infty)$) because squaring any number always gives a positive or zero result.

We can **represent a function** in math by the three methods,

Set of Ordered Pairs	For instance, for a function, $f(x) = x^3$ The set of ordered pairs is: $f = \{(1, 1), (2, 8), (3, 27)\}$												
Tabular Format	<table><tr><td>Input</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td></tr><tr><td>Output</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr></table>	Input	-2	-1	0	1	2	Output	0	1	2	3	4
Input	-2	-1	0	1	2								
Output	0	1	2	3	4								
Graphical Format	• The function $f(x) = x^2$ produces a parabola. • The function $f(x) = x^3$ results in a cubic curve.												

	For $f(x) = x + 1$ and $g(x) = x - 2$, what is $(f \cdot g)(x)$?	Given functions are $f(x) = x + 1$ and $g(x) = x - 2$ We need to calculate $(f \cdot g)(x)$, $(f \cdot g)(x) = f(x) \cdot g(x)$ $(f \cdot g)(x) = (x + 1)(x - 2)$ $(f \cdot g)(x) = x^2 - 2x + x - 2$ $(f \cdot g)(x) = x^2 - x - 2$ So, $(f \cdot g)(x) = x^2 - x - 2$.
	If $f(x) = 5x - 2$, find $f^{-1}(x)$.	Given function is $f(x) = 5x - 2$ Let $y = 5x - 2$ Now, solve for x in terms of y: $x = 5y - 2$ Now solve for y: $x + 2 = 5y \Rightarrow y = (x + 2)/5$ Therefore, $f^{-1}(x) = (x + 2)/5$

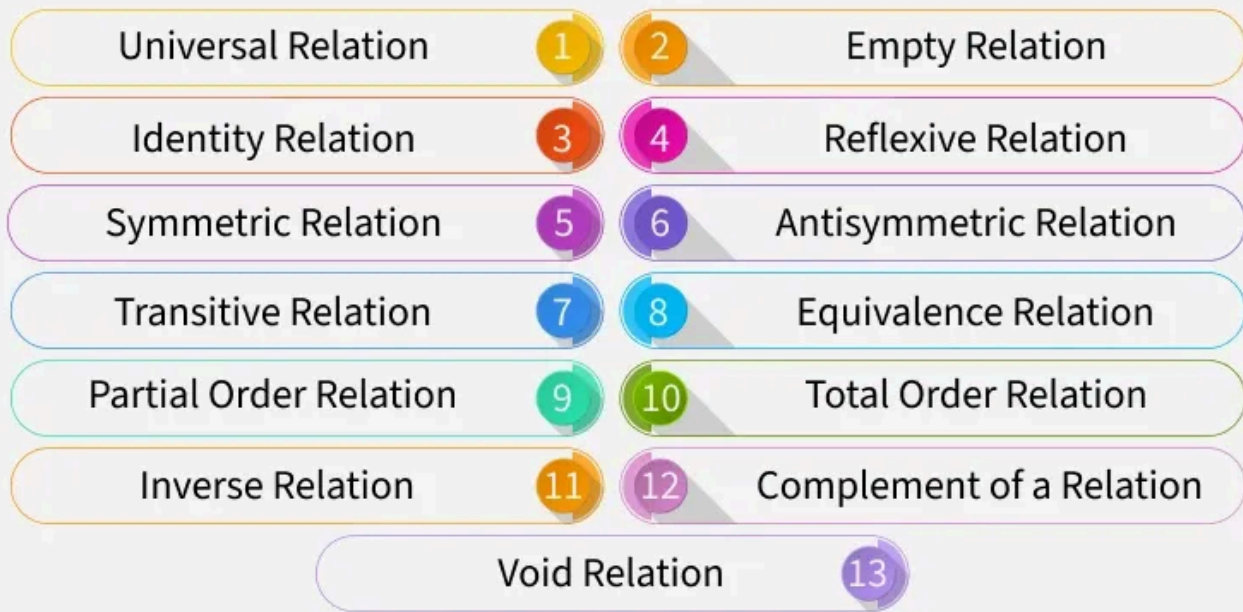
A **relation** in mathematics is defined as the relationship between two sets.

If we are given two sets, set A and set B, and set A has a relation with set B, then each value of set A is related to a value of set B through some unique relation. Here, set A is called the domain of the relation, and set B is called the range of the relation.

Representation of Relations

Set Builder Notation	For example, if we are given two sets, set $X = \{2, 4, 6\}$ and set $Y = \{4, 8, 12\}$. Then, on observing clearly, we can see that each element of set Y is twice each element of set X the relation between them is, $R = \{(a, b) : b \text{ is twice of } a, a \in X, b \in Y\}$
Roaster Form	For example, if we are given two sets, set $X = \{2, 4, 6\}$ and set $Y = \{4, 8, 12\}$. Then the relation between set X and set Y is represented using the relation R such that, $R = \{(2, 4), (4, 8), (6, 12)\}$

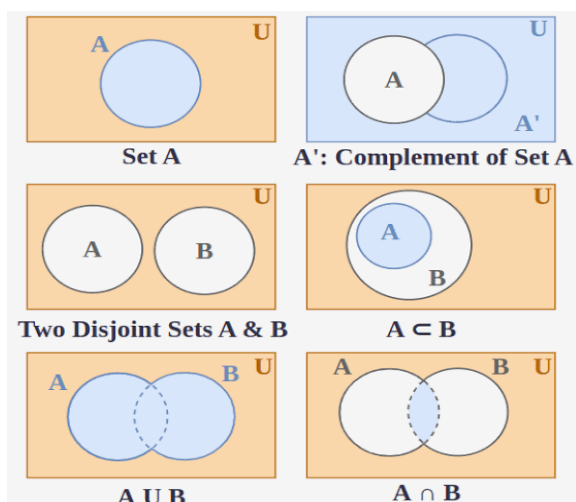
Types of Relations in Maths



Reflexive Relation	A relation R on a set A is a reflexive relation if every element is related to itself. That means $(a, a) \in R$ for every $a \in A$. Example: For $A = \{1, 2\}$, $R = \{(1, 1), (2, 2), (1, 2)\}$. then R is reflexive because each element (1 and 2) is related to itself.
Symmetric Relation	A relation R is a symmetric relation if whenever $(a, b) \in R$, then $(b, a) \in R$ as well. In simple words, if one element is related to another, the second is also related to the first. Example: If $R = \{(1, 2), (2, 1)\}$, R is symmetric. then R is symmetric because 1 is related to 2 and 2 is related to 1.
Anti Symmetric Relation	A relation R is anti-symmetric if whenever $(a, b) \in R$ and $(b, a) \in R$, then $a = b$. That means both (a, b) and (b, a) can exist only if they are the same element. Example: If $R = \{(1, 2)\}$, R is anti-symmetric because there is no pair $(2, 1)$ where $a \neq b$.
Transitive Relation	A relation R is a transitive relation if whenever $(a, b) \in R$ and $(b, c) \in R$. This means the relation “passes through” — if a is related to b and b is related to c , then a must be related to c . Example: If $R = \{(1, 2), (2, 3), (1, 3)\}$, R is transitive because $1 \rightarrow 2$ and $2 \rightarrow 3$ implies $1 \rightarrow 3$.
Equivalence	A relation is called an equivalence relation if it is reflexive, symmetric, and transitive —

Relation	all three at once. Equivalence relations group elements into “classes” of items that are equivalent in some way. Example: The relation “has the same age” among people is an equivalence relation, since it’s reflexive (a person has the same age as themselves), symmetric (if A has the same age as B, then B has the same age as A), and transitive (if A and B have the same age, and B and C have the same age, then A and C have the same age).
Inverse Relation	The inverse of a relation reverses the order of every pair. If $(a, b) \in R$, then $(b, a) \in R^{-1}$. Example: If $R = \{(1, 2), (3, 4)\}$, then $R^{-1} = \{(2, 1), (4, 3)\}$.

Set Theory



Set Identities	
Identity	Name
$A \cap U = A$ $A \cup \emptyset = A$	Identity laws
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws
$\overline{(\overline{A})} = A$	Complementation law
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive laws
$\overline{A \cap B} = \overline{A} \cup \overline{B}$ $\overline{A \cup B} = \overline{A} \cap \overline{B}$	De Morgan's laws
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption laws
$A \cup \overline{A} = U$ $A \cap \overline{A} = \emptyset$	Complement laws

@mathsbook7474

Statistics

Problem - Find the mean, median, mode, and range for the given data

190, 153, 168, 179, 194, 153, 165, 187, 190, 170, 165, 189, 185, 153, 147, 161, 127, 180

Solution:

For Mean: Number of observations = 18

$$\begin{aligned}\text{Mean} &= (\text{Sum of observations}) / (\text{Number of observations}) = \\ &= (190+153+168+179+194+153+165+187+190+170+165+189+185+153+147 \\ &+161+127+180) / 18 \\ &= 2871/18 \\ &= 159.5\end{aligned}$$

For Median: The ascending order of given observations is,

127, 147, 153, 153, 153, 161, 165, 165, 168, 170, 179, 180, 185, 187, 189, 190, 190, 194

Here, $n = 18$

$$\begin{aligned}\text{Median} &= 1/2 [(n/2) + (n/2 + 1)]^{\text{th}} \text{ observation} \\ &= 1/2 [9 + 10]^{\text{th}} \text{ observation} \\ &= 1/2 (168 + 170) \\ &= 338/2 \\ &= 169\end{aligned}$$

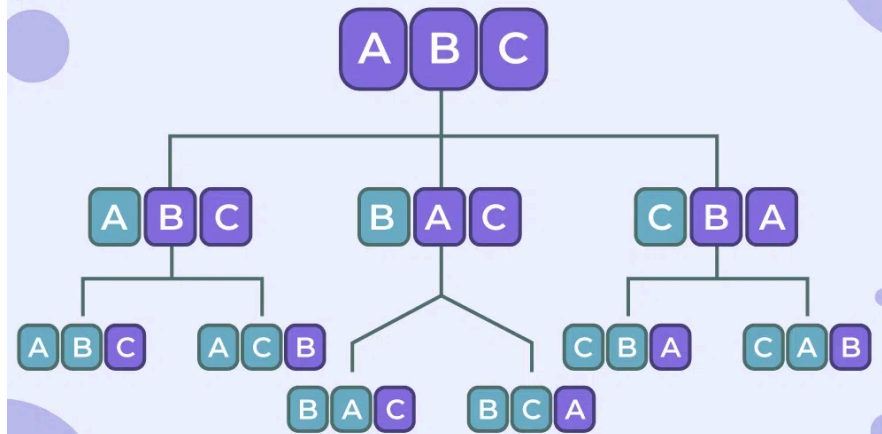
For Mode:

The number with the highest frequency, which is 3 belongs to 153

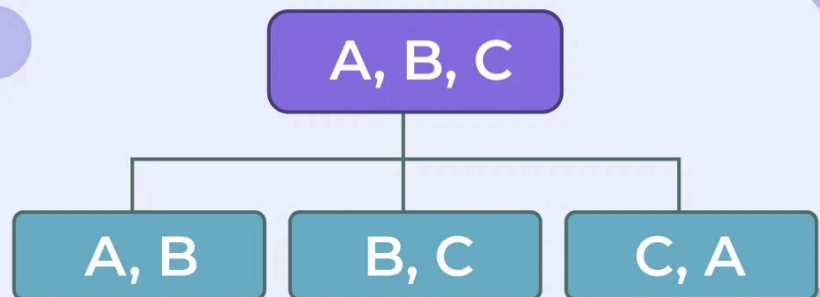
Thus, mode = 153

Permutation and Combination

$${}^n P_r = \frac{n!}{(n-r)!}$$



$${}^n C_r = \frac{n!}{r!(n-r)!}$$



If ${}^6 P_r = 360$ and If ${}^6 C_r = 15$, find r
[47th BCS General]

$$\begin{aligned} {}^n P_r &= {}^n C_r \times r! \\ \Rightarrow 360 &= 15 \times r! \\ \Rightarrow r! &= 360 / 15 = 24 \end{aligned}$$

How many ways can 4 prizes be given away to 3 boys, if each boy is eligible for all the prizes?

Let the 3 boys be B1, B2, B3 and 4 prizes be P1, P2, P3 and P4
Now B1 is eligible to receive any of the 4 available prizes (so 4 ways)
B2 will receive prize from rest 3 available prizes (so 3 ways)
B3 will receive his prize from the rest 2 prizes available (so 2 ways)
So total ways would be: $4 \times 3 \times 2 \times 1 = 24$ Ways

How many different words can be formed by the word **ALLAHABAD**?

- a) keeping the vowels in even place
b) both L do not appear together

ALLAHABAD has 9 letters, out of these 9 letters there are 4 A's and 2 L's, so, total permutations = $9! / 4! \cdot 2! = 7560$ (Ans)

a) There are 4 vowels, which are A and have 4 even places. These even places can be occupied by 4 vowels. In $4! / 4! = 1$ way.

	2		4		6		8	
--	---	--	---	--	---	--	---	--

In other five places 5 other letters can be occupied of which there are 2 L's. Number of ways = $5! / 2!$
= 60 ways.

So total words = $60 \times 1 = 60$ ways (Ans)

b) Taking both L's together and treating them as one letter we have 8 letters to count, out of which A repeats 4 times. So, These 8 letters can be arranged in $8! / 4! = 1680$ ways.

	Also two L can be arranged themselves in $2!$ ways. So, Total no. of ways in which L are together = $1680 \times 2 = 3360$ ways. Now, Total arrangement in which L never occur together = Total arrangement - Total no. of ways in which L occur together. = $7560 - 3360$ = 4200 ways (Ans)
In how many ways can 10 examination papers be arranged so that the best and the worst papers never come together?	10 papers can be arranged in $10!$ Ways. Keeping the worst and best paper as one paper, total 9 papers can be arranged in $9!$ Ways, while the worst and best paper can be arranged in $2!$ Ways. So total $9! * 2!$ Ways Total papers as per requirement = $10! - (9! * 2!)$ = $9! (10 - 2) = 8 * 9!$ (Ans)
A two member committee comprising of one male and one female member is to be constitute out of five males and three females. Amongst the females. Ms. A refuses to be a member of the committee in which Mr. B is taken as the member. In how many different ways can the committee be constituted ?	${}^5C_1 \times {}^3C_1 - 1$ = $(5 * 3) - 1$ = $15 - 1$ = 14 (Ans)
a) Find the number of triangles which can be formed by joining the angular points of a polygon of 8 sides as vertices. b) There are 10 points in a plane out of which 4 are collinear. Find the number of triangles formed by the points as vertices.	a) The polygon has 8 angular points while the triangle has 3. So, total triangles can be = 8C_3 (Ans) b) Triangles can be made from 10 points are ${}^{10}C_3$, while triangles can be made from 4 collinear points are 4C_3. As 4 collinear points are, so total triangles will be = ${}^{10}C_3 - {}^4C_3 = 120 - 4 = 116$ (Ans)
c) In a party every person shakes hands with every other person. If there are 105 hands shakes, find the number of person in the party.	c) ${}^nC_2 = 105$ $\Rightarrow n! / \{ (n-2)! * 2! \} = 105$ $\Rightarrow n(n-1) / 2 = 105$ $\Rightarrow n^2 - n = 210 \Rightarrow n = 15, -14$, so $n = 15$
Three gentlemen and three ladies are candidates for two vacancies. A voter has to vote for two candidates. In how many ways can one cast his vote?	There are 6 candidates and a voter has to vote for any two of them. So, the required number of ways = ${}^6C_2 = 6! / 2! \times 4! = 6 * 5 / 2 = 15$
A question paper has two parts, A and B, each containing 10 questions. If a student has to choose 8 from part A and 5 from part B, in how many ways can he choose the questions?	From part A, out of 10 questions we have to choose 8, which is ${}^{10}C_8$ and in part B, we have to choose 5 out of 10, which is ${}^{10}C_5$ So total ways = ${}^{10}C_8 * {}^{10}C_5 = 11340$

The number of positive integers which can be formed by using any number of digits from 0, 1, 2, 3, 4, 5 without repetition	সংখ্যার ধরন	গঠনের উপায় (Calculation)	মোট সংখ্যা
	১-অঙ্কের	1, 2, 3, 4, 5 (০ বাদে)	5
	২-অঙ্কের	প্রথম ঘরে ৫টি অপশন (০ বাদে), ২য় ঘরে ৫টি। (5×5)	25
	৩-অঙ্কের	$5 \times 5 \times 4$	100
	৪-অঙ্কের	$5 \times 5 \times 4 \times 3$	300
	৫-অঙ্কের	$5 \times 5 \times 4 \times 3 \times 2$	600
	৬-অঙ্কের	$5 \times 5 \times 4 \times 3 \times 2 \times 1$	600
Total numbers = $5 + 25 + 100 + 300 + 600 + 600 = 1630$			
In the next World cup of cricket there will be 12 teams, divided equally in two groups. Teams of each group will play a match against each other. From each group 3 top teams will qualify for the next round. In this round each team will play against each others once. Four top teams of this round will qualify for the semifinal round, where they play the best of three matches. The Minimum number of matches in the next World cup will be	The number of matches in first round = ${}^6C_2 + {}^6C_2$ Number of matches in next round = 6C_2 Number of matches in semifinals = 4C_2 Total number of matches = ${}^6C_2 + {}^6C_2 + {}^6C_2 + {}^4C_2 + 2 = 53$		
There are 10 person among whom two are brother. The total number of ways in which these persons can be seated around a round table so that exactly one person sit between the brothers , is equal to:	Among 2 brothers, in a circular table, they can sit in 2 ways, ($B_1 - A - B_2$) and ($B_2 - A - B_1$) Rest 8 places, other 8 persons can be sit in a circular table = $(8 - 1)! = 7!$ Ways. So total ways = $2 * 7!$		
If letters of the work KUBER are written in all possible orders and arranged as in a dictionary, then the rank of the word KUBER will be - b) LABOUR - 242	Step 1: Arrange the letters in alphabetical order - B, E, K, R, U Step 2: Count words before "KUBER" Words starting with 'B' = $4! = 24$ Words starting with 'E' = $4! = 24$ Words starting with 'K' and second letter 'B' = $3! = 6$ Words starting with 'K' and second letter 'E' = $3! = 6$ Words starting with 'K' and second letter 'R' = $3! = 6$ Words starting with 'K', second letter 'U', third letter 'B', forth letter 'E' and then last word 'R' After 24 (B) + 24 (E) + 6 (KB) + 6 (KE) + 6 (KR) = 66 words, the next word is KUBER. Rank of "KUBER" = $66 + 1 = 67$		
10 students are to be seated in two rows equally for the Mock test in a room. There are two sets of papers, Code A and Code B. each of two	Let's choose 5 students for the first row, which can be done in $5!$ ways, and rest 5 for the second row, can be done in $5!$ Ways. So, students of each row can be arranged as = $5! \times 5!$ ways Among 10 students 5 can be seated in ${}^{10}C_5$ ways.		

rows can have only one set of paper but different that from other row. In how many ways these students can be arranged ?	Two sets of paper can be arranged themselves in = $2!$ ways Thus, Total arrangement = ${}^{10}C_5 \times 5! \times 5! \times 2!$ = 7257600
Find the number of ways in which 8064 can be resolved as the product of two factors?	$8064 = 2^7 \times 3^2 \times 7$ Total factors = $(7 + 1) \times (2 + 1) \times (1 + 1) = 8 \times 3 \times 2 = 48$ Number of ways = $48 / 2 = 24$ (Ans)
Ten different letters of alphabet are given, words with 5 letters are formed from these given letters. Then, the number of words which have at least one letter repeated is	Number of words which have at least one letter repeated = Total words - Words with no repetition = $10^5 - {}^{10}P_5$ = $100000 - 30240$ = 69760

Probability

Tickets numbered 1 to 20 are mixed up and then a ticket is drawn at random. What is the probability that the ticket drawn has a number which is a multiple of 3 or 5?	Here, $S = \{1, 2, 3, 4, \dots, 19, 20\}$ Let E = event of getting a multiple of 3 or 5 = $\{3, 6, 9, 12, 15, 18, 5, 10, 20\}$ $\therefore P(E) = n(E) / n(S) = 9 / 20$
A bag contains 2 red, 3 green and 2 blue balls. Two balls are drawn at random. What is the probability that none of the balls drawn is blue?	Total balls $(2 + 2 + 3) = 7$, out of which 2 balls can be drawn in ${}^7C_2 = 21$ ways. Mentioned that none of the balls will be blue, so we have to count only red and green balls, which is $2+3 = 5$ ball, that is ${}^5C_2 = 10$ ways So probability = $10 / 21$
In a box, there are 8 red, 7 blue and 6 green balls. One ball is picked up randomly. What is the probability that it is neither red nor green?	Total balls = $8 + 7 + 6 = 21$ One ball will be neither red nor green, so it will be blue ball, which is one ball out of 7. Probability = $7 / 21 = 1 / 3$
Two dice are thrown simultaneously. What is the probability of getting two numbers whose product is even?	In a simultaneous throw of two dice, we have $n(S) = (6 \times 6) = 36$ Then, $E = \{(1, 2), (1, 4), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 2), (3, 4), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (5, 2), (5, 4), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\} = 27$ $\therefore n(E) = 27$ $\therefore P(E) = n(E) / n(S) = 27 / 36 = 3 / 4$
In a class, there are 15 boys and 10 girls. Three students are selected at random. The probability that 1 girl and 2 boys are selected, is:	Total students = $15 + 10 = 25$ out of which 3 students will be selected, can be done in ${}^{25}C_3 = 2300$ ways i.e. $n(S)$ We can select 1 girl out of 10 in ${}^{10}C_1 = 10$ ways and 2 boys out of 15 in ${}^{15}C_2 = 105$ ways. So probability = $105 \times 10 / 2300 = 21 / 46$
Two dice are tossed. The probability that the total score is a prime number is:	In a simultaneous throw of two dice, we have $n(S) = (6 \times 6) = 36$ Then, $E = \{(1, 1), (1, 2), (1, 4), (1, 6), (2, 1), (2, 3), (2, 5), (3, 2), (3, 4), (4, 1), (4, 3), (5, 2), (5, 6), (6, 1), (6, 5)\} = 15$ $\therefore n(E) = 15$

	$\therefore P(E) = n(E) / n(S) = 15 / 36 = 5 / 12$
A bag contains 4 white, 5 red and 6 blue balls. Three balls are drawn at random from the bag. The probability that all of them are red, is:	Probability = $n(S) / n(E) = {}^{15}C_3 / {}^5C_3 = 10 / 455$
An urn contains 6 red, 4 blue, 2 green and 3 yellow marbles. If 4 marbles are picked up at random, what is the probability that at least one of them is blue?	Here, 4 ball have to select out of $(6+4+2+3) = 15$ balls, so $n(S) = {}^{15}C_4 = 1365$ Let, E = drawing 4 balls where none is blue = drawing 4 balls out of 11 remaining balls = ${}^{11}C_4 = 330$ So probability of drawing 4 balls that are not blue = $330 / 1365$ Finally, probability of drawing 4 balls where at least one ball is blue = $1 - (330/1365) = 1065 / 1365$
Server X and Server Y, to distribute incoming web traffic. Server X handles 60% of the total incoming requests, while Server Y processes the remaining network load. Due to varying hardware conditions, the probability of a network request experiencing high latency is 2% when processed by Server X, and 4% when routed through Server Y. Based on these performance metrics, determine the total probability that any randomly selected user request will encounter high latency, and calculate the conditional probability that a delayed request was specifically handled by Server X. [ISTCL AP - 2026]	$P(X) = 0.60$ (Server X handles 60% of traffic) $P(Y) = 0.40$ (Server Y handles the remaining 40%) $P(L X) = 0.02$ (Latency probability given Server X) $P(L Y) = 0.04$ (Latency probability given Server Y) Total Probability of High Latency, $P(L) = P(X) * P(L X) + P(Y) * P(L Y)$ = $0.60 * 0.02 + 0.40 * 0.04$ = 0.028 Conditional Probability that a Delayed Request was Handled by Server X, $P(X L) = P(X) * P(L X) / P(L)$ = $0.60 * 0.02 / 0.028$ = $3 / 7$